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Credit risk prediction with machine learning



A seven-input model with AI-assisted explanations

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Course

Introduction to AI in Business and Management

Professor

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Understanding Credit Risk

Jan Jurczyk

Credit risk is the risk that a borrower will fail to meet their repayment obligations.

How banks assess credit risk?

Probability of Default
How likely is default?
01

Loss Given Default
How much could be lost?
02

Exposure at Default
How much is owed at default?
03

Two possible classification errors

False Approval
Risky applicant classified as safe



False Rejection
Reliable applicant classified as risky

How Banks Assess Credit Risk

Research shows that more complex models can improve prediction, but performance and explainability must be evaluated.

Manual Assessment
Analyst reviews applicant information

1

Rule-Based Systems
Predefined lending criteria

2

Statistical Models
Logistic regression

3

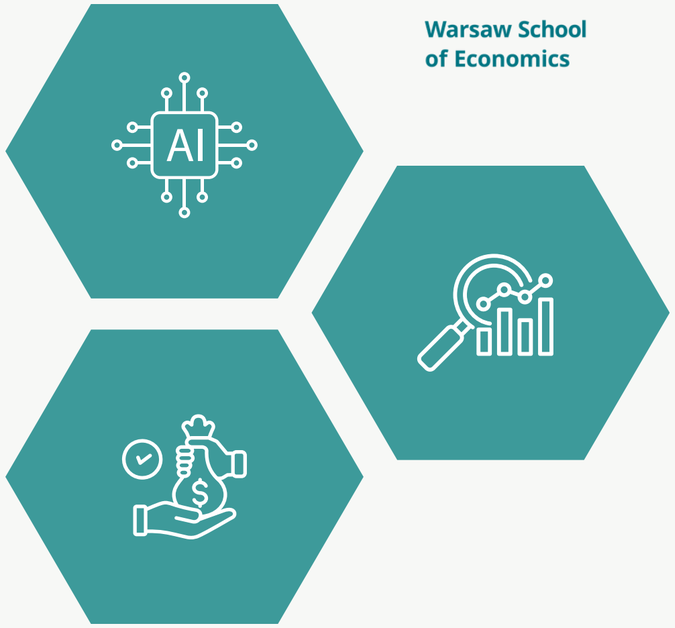
Machine Learning
Random forest and other algorithms

4

Source: Hand & Henley (1997); Lessmann et al. (2015)

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How well can a simple machine learning model classify credit risk using seven applicant and loan characteristics?



What do we want to find out?

Can machine learning provide useful risk classifications compared with a simple baseline?



UCI German Credit dataset

Dataset size	**1,000 historical cases**
Class distribution	**70% Good · 30% Bad**
Selected inputs	**7 of 20 available features**
Target	**Good / Bad risk**

Seven model inputs

- Credit amount
- Loan duration
- Loan purpose
- Checking account status
- Savings
- Employment duration
- Credit history

Historical risk labels; amounts recorded in Deutsche Mark (DM)

Source: Hofmann (1994)

Training, validation and testing

Nicolai Hönnige

01 → 02 → 03

Development data

800 cases
Five-fold cross-validation
640 training + 160 validation cases per fold



Training learns patterns

Model selection

Compare three models
Select using balanced accuracy
Fix the model and decision threshold



Validation compares models

Final test

200 separate cases
Evaluate the selected model once
No further tuning on test results



Testing checks performance on unseen cases

Source: Scikit-learn developers (n.d.-b, n.d.-c)

How we teach and check the models

Nicolai Hönnige

Learning: 640 cases

Person 189 from 640



Applicant information

credit history
amount
....



Good risk

Person 3 from 640



Applicant information

credit length
duration
....



Bad risk

Logistic Regression

Random forest

The models **learn patterns from examples** with known **labels**.

Checking: 160 other cases

Person 12 from other 160



Applicant information

credit length
duration
....



Risk hidden for model



Baseline: Good risk

Logistic Regression: Bad risk TRUE

Random forest: Good risk

We **check all 160 cases**, repeat this **across five folds** and compare average scores.

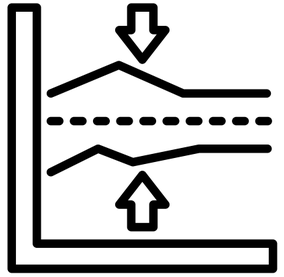
The models we compared

Nicolai Hönnige

Always-Good baseline

Predicts Good risk for **every** case

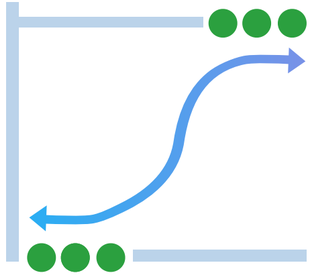
A **simple reference** for **comparison**



Logistic regression

Combines **weighted inputs** into a score

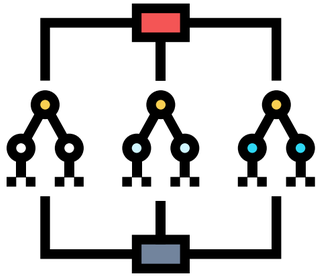
A relatively **easy model** to **interpret**



Random forest

Combines predictions from many **decision trees**

Can capture more **complex relationships**



Main selection criterion: Balanced accuracy

Average of the detection rates for Good and Bad risks

Source: Scikit-learn developers (n.d.-a, n.d.-d, n.d.-e, n.d.-f)

Machine learning predicts the class.

Generative AI explains the result.

Prompt summaries

Simple prompt	“Explain the results to a bank manager.”
Structured prompt	“Explain both error types, include the numbers and denominators, and state the limitations.”

Same data per scenario · 120–180-word target for both



What we observed

- 3 scenarios × 2 prompts × 5 responses = 30 responses
- One actual-results scenario and two synthetic scenarios
- Complete counts and denominators: **Simple 10/15 · Structured 15/15**
- Within the word limit: **Simple 15/15 · Structured 12/15**
- Structured was more complete, but did not always follow the word limit.

Own experiment using OpenAI GPT-5.4 mini; AI-assisted assessment reviewed by Nicolai Hönnige.

From model to business value

Philipp Pehlemann

Jan | Nicolai | **Philipp** | Zosia

1	2	3	4	5	**6**	**7**	**8**	9
Research question	Choose data	Exploratory data analysis	Building the models	Validation and model selection	Final test and results	Interpretation	Costs and benefits	Risks and Conclusion

How good is the model, and is it worth it?

74%
balanced accuracy on the final test

46/60
bad risks found (40/140 false alarms)

0.55
assumed cost per case (baseline 1.5)

A simple model beats the baseline

Philipp Pehlemann

Validation: 800 development cases, 5 folds

Model	Balanced accuracy	Bad-risk recall	ROC-AUC
Always-good baseline	50.00%	0.00%	0.5
Logistic regression	70.90%	71.30%	0.77
Random forest	70.20%	70.00%	0.78

- Both models clearly beat the baseline and are almost identical
- We choose the simpler one - logistic regression

Final test: 200 unseen cases

73.0%

accuracy baseline is 70.0%

74.0%

balanced accuracy baseline is 50.0%

0.81

ROC-AUC baseline is 0.50

- Model frozen before the test, evaluated once
- Slightly above validation: one random split, no proof of improvement

Accuracy alone misleads: the baseline reaches 70% without finding a single bad risk

Most bad risks are found

Philipp Pehlemann

- False approval: bad risk passes as good → possible loan loss (5x cost
- False rejection: good applicants flagged as bad → lost margin and customer (1x cost)

- 46 of 60 bad risks detected (14 missed)
- 40 of 140 good applicants wrongly flagged
- 46 of 86 flagged cases truly bad (precision 53.5%)

	Predicted “good risk”	**Predicted “bad risk”**
Actually “good risk”	Correct approval	False rejection
Actually “bad risk”	False approval	Correct rejection (bad risk detected)

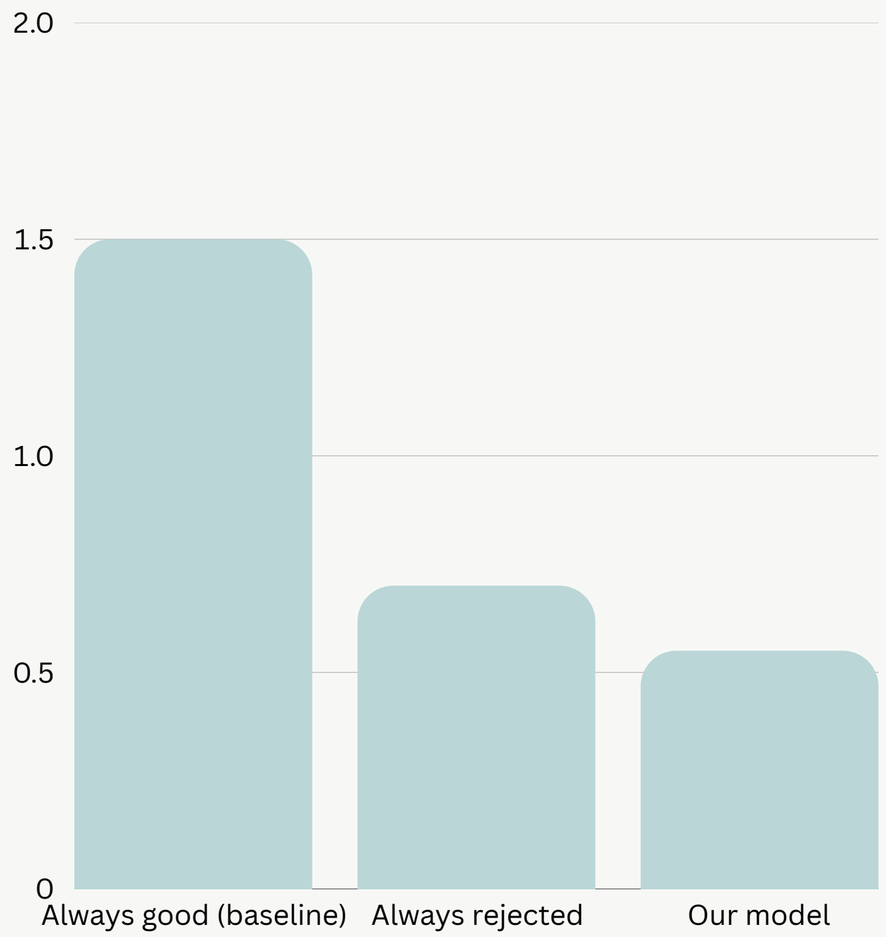
	Predicted “good risk”	**Predicted “bad risk”**
Actually “good risk”	100	40
Actually “bad risk”	14	46

Catching more bad risks means rejecting more good applicants

Lower assumed error costs

Philipp Pehlemann

Assumed error cost per case (units, not euros)



5 units per false approval, 1 per false rejection; 200 test cases



Implementation costs

- One-off: data, development, integration, compliance
- Recurring: hosting, monitoring, retraining
- Potential cost driver: manual review of flagged cases, 43% in our test set.



Benefits

- 76.7% of bad risks detected (baseline: none)
- Consistent, reproducible first screening



Not shown

- Cheapest option only if a false approval costs about 0.9–7× a false rejection
- Profitability needs the bank’s own loss, margin and review costs

Zofia Wiśniewska

Data limitations

73%	74%
Model Accuracy	Balance Accuracy

76.7%	53.5%
Bad-risk recall	Bad-risk precision

- **False negatives (14)** → risky borrowers incorrectly classified as good - potential financial losses.
- **False positives (40)** → creditworthy borrowers incorrectly classified as risky may lead to unfair loan rejections.
- **Limited dataset** → only 1,000 historical observations.
- **Limited features** → only 7 selected variables out of 20 original features.
- **Generalisation** → historical data may not represent current borrowers or economic conditions.

Risks

Ethical

- **Algorithmic bias** → indirect discrimination through socioeconomic variables.
- **Lack of transparency** → unclear model predictions.
- GenAI **hallucinations** → potentially inaccurate AI-generated explanations.

Social

- Automated decisions can **normalise unfair systems**
- Financial **exclusion** & risky applicants
- **Social inequality** → reinforcement of existing socioeconomic inequalities.
- **Loss of trust** → reduced confidence in automated financial decisions.

Legal

- **EU AI Act** → credit scoring is classified as high-risk AI.
- **GDPR** → requirements for automated decision-making and data protection.
- **Data privacy** → sharing applicant data with external APIs.

Source: EU Artificial Intelligence Act (2024).

Zofia Wiśniewska

Protective measures

Fairness safeguards

- Tests of the model's performance across groups
- Identification of bias in training data

Privacy safeguards

- Data minimization and anonymization
- Restricted access and storage
- Data protection review

Transparency safeguards

- Document model purpose + limitations
- Clear explanations for affected users
- Not presenting predictions as certainty

Monitoring & Accountability

- Evaluation of performance over time
- Incident response process

Legal support

- Data Protection Impact Assessment
- Clear accountability for model owners

Source: European Commission. (2019)

Conclusion



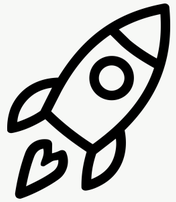
What did we achieve?

- Developed a machine-learning credit-scoring prototype.
- Achieved **73% accuracy** and a **ROC-AUC of 0.810**.
- Integrated GenAI to translate model outputs into understandable explanations.



What are the main limitations?

- Small and outdated dataset.
- Prediction errors with potentially serious financial and social consequences.
- Real-world reliability have not been fully validated.



What comes next?

- Use larger, more representative, and up-to-date datasets.
- Improve model validation and fairness testing.
- Introduce human supervision.

Live-Demo

Enter a fictional profile, explore the predicted risk class and see how AI explains the result.



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